

CYBER SECURITY



Security Framework Er. Prakash Chandra Adhikari

What Is a Security Framework?

- A security framework is a structured approach to managing cyber security
- It provides policies, procedures, and controls
- Frameworks help organisations manage and reduce security risk

Why Security Frameworks Are Important

- They provide consistency in security practices
- Support compliance with legal and regulatory requirements
- Help organisations identify and manage risks systematically

Key Components of a Security Framework

- Security policies define rules and expectations
- Standards and guidelines support policy implementation
- Controls enforce protection mechanisms

Security Governance

- Governance defines responsibility and accountability for security
- Senior management oversight is essential
- Clear roles improve decision-making and risk ownership

Risk Management Fundamentals

- Risk is the likelihood of a threat exploiting a vulnerability
- Risk management identifies, analyses, and mitigates risks
- Frameworks embed risk management into security planning

Risk Assessment Process

- Identify assets and threats
- Assess vulnerabilities and potential impact
- Prioritise risks based on likelihood and severity

Security Policies

- Policies define acceptable and unacceptable behaviour
- They guide employees and system users
- Policies must be clear, enforceable, and reviewed regularly

Security Controls

- Technical controls include firewalls and encryption
- Administrative controls include policies and training
- Physical controls protect facilities and hardware

NIST Cybersecurity Framework

- Widely used international security framework
- Five core functions: Identify, Protect, Detect, Respond, Recover
- Supports risk-based security management

ISO/IEC 27001 Framework

- International standard for information security management
- Focuses on Information Security Management Systems (ISMS)
- Emphasises continual improvement

Other Security Frameworks

- COBIT focuses on IT governance
- PCI DSS addresses payment card security
- Framework selection depends on organisational needs

Applying Frameworks in Practice

- Frameworks must be tailored to the organisation
- Risk appetite and business context are critical
- Continuous monitoring and improvement are required

Challenges in Using Frameworks

- Complexity and cost of implementation
- Lack of organisational support
- Keeping frameworks updated with evolving threats